Earths angular momentum: $I = \frac{2}{5}mr^2$

$L = I\omega$ $\quad m = 6\times10^{24}$ kg $\quad r = 6\times10^{6}$ m $\quad \omega = \frac{v_t}{r} = \frac{2\pi}{T}$

$$L = \left(\frac{2}{5}mr^2\right)\left(\frac{2\pi}{T}\right)$$

★ if stupendous man stops rotation in 100s

$$\tau = \frac{dL}{dt} = rF_\perp = \frac{\Delta L}{\Delta t} \rightarrow F_\perp = \frac{L_f - L_i}{t}\cdot\frac{1}{r}$$

$$\tau = rF_\perp$$

$L_f = 0 \quad L_i = 6\times10^{33}\ kg\,m^2/s \quad r = 6\times10^6 m \quad t = 100s$

$$F_\perp = \frac{0 - 6\times10^{33} kg\,m^2/s}{10^2}\cdot\frac{1}{6\times10^6 m} = \frac{-10^{33}}{10^8} = -10^{25} N$$

$M = 200$ kg $\quad R = 1.5$m $\quad m = 70$ kg $\quad I = \frac{MR^2}{2}$

$v_B = 380$ m/s

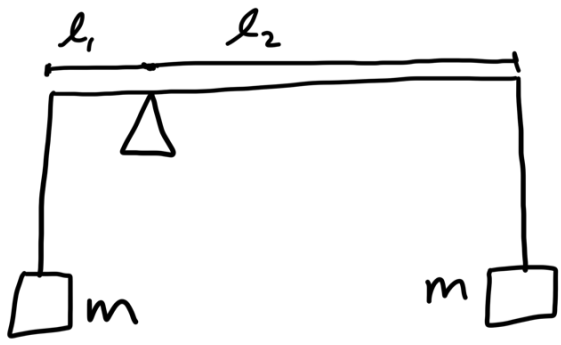
$L = I\omega$

$$L_i = L_f$$

$$0 = -mv_BR + I\omega_f$$

$$\omega_f = \frac{mv_BR}{\frac{MR^2}{2}} = \frac{mv_B}{MR/2} = \frac{(70kg)(380m/s)}{(200kg)(1.5m)/2} =$$

huh?



$h = 3v^2/4g$

$$E_i = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = E_f = mgh$$

$K_{rot} = \frac{1}{2}I\omega^2$

$$\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{mg\,3v^2}{4g}$$

$$\frac{1}{2}mv^2 + \frac{1}{2}I\frac{v^2}{R^2} = \frac{m3v^2}{4}$$

$$\frac{1}{2}I\frac{v^2}{R^2} = \frac{3mv^2}{4} - \frac{1}{2}mv^2$$

$$\frac{1}{2}I\frac{v^2}{R^2} = \frac{1}{4}mv^2 \qquad \boxed{I = \frac{1}{2}mR^2}$$